

CSI243 FUNCTIONAL PROGRAMMING

Lecturer **Name :** **TEBOGO SEIPONE**
Email Address: seiponet@ub.ac.bw
Office Number: Block 247 Room 284

Credits: 3

Pre-requisites: CSI142

Co-requisites: None

Aims and Learning Objectives

The course aims to introduce Functional programming as one of the models in Computer Science. The course introduces the main concepts behind the functional programming paradigm and problem solving using functional languages. It emphasizes important concepts of functional programming such as recursive definitions, higher-order functions, and type inference. Programming exercises will illustrate the utility of list-processing, pattern matching, abstraction of data/control, strong typing, and parameterized modules (functors). This course also emphasizes mathematical reasoning involved in the design of functional programs and techniques for proving properties about functions so defined.

Learning Outcomes

Upon successful completion of the course students should be able to:

- Explain the principles and techniques of programming with functions
- Explain the main features of a modern functional language
- Write non-trivial functional programs
- Develop applications using functional programming languages

Topics

- Define functional programming, and compare to imperative programming languages
- FP concepts
 - recursive definitions
 - higher-order functions
 - type inference
 - polymorphism
 - abstract data types
 - modules
 - Definitions of functions
- Mathematical reasoning in the design of functional programs
- List-processing, pattern matching
- Abstraction of data/control, strong typing, and parameterized modules (functors)
- Define polymorphism; abstract data types; type checking; type inference
- Overview of complexity of functions
- Design, implement, and test applications using Haskell as a functional programming language

Lecture: Tuesday 1200 – 1400 hours (Block 247 Room 035)

Lab: Thursday 1100 – 1300 hours (Block 247 Labs 292 and 293)

Office Hour: Thursday 1400 – 1500 hours

Modes of Assessment

1. **CA** (50% made up of 2 Tests, Weekly Lab work)
 - **Test 1:** Thursday 15th September, 1100 hours
 - **Test 2:** Thursday 3rd November, 1100 hours
2. **Examination** (50%) – Date and time based on the UB timetable

Reading (and other resources) list

1. Simon Thompson, Haskell: The Craft of Functional Programming, Pearson, 3 ed, 2011
2. Haskell support <https://www.haskell.org>
3. Miran Lipovaca, Learn You a Haskell for Great Good!: A Beginner's Guide, 1st Edition, Kindle Edition Online at learnyouahaskell.com
4. Fernández M, Models of computation: An introduction to computability theory, Springer, 2009
5. Ederton H B, A mathematical introduction to logic, Harcourt/Academic Press, 2ed, 2001
6. Other relevant reading material

Class Policies

- Class notes, syllabus, announcements, and other related materials can be accessed on MOODLE <http://www.cs.ub.bw/moodle/>
Please make sure that you regularly check this address for announcements and course related materials
- When in class, please turn off all pagers, phones, and beepers
- Projects and assignments must be individual unless stated otherwise
- Any test/assignment/quiz/lab not done will be marked 0 unless there is a genuine reason, provided in reasonable time frame, for being absent, in which case a makeup assessment will have to be done

Academic Dishonesty

- Harms the good reputation of our University
- Affects your personal integrity
- Is a barrier to the acquisition of the knowledge, skills and attitudes you would have acquired at the end of your study
- Decreases your value in the eyes of employers
- Has a heavy penalty including expulsion from UB

DO NOT CHEAT OR CONDONE CHEATING BY OTHERS